



Clinical RAG — Multi-Agent Search Over Medical Guidelines

clinic-rag.vercel.app
github.com/waseem253/clinic-rag-agent

HIPAA-aware multi-step agent · per-page citations ·
1,614 indexed chunks · 5-day solo build

The problem

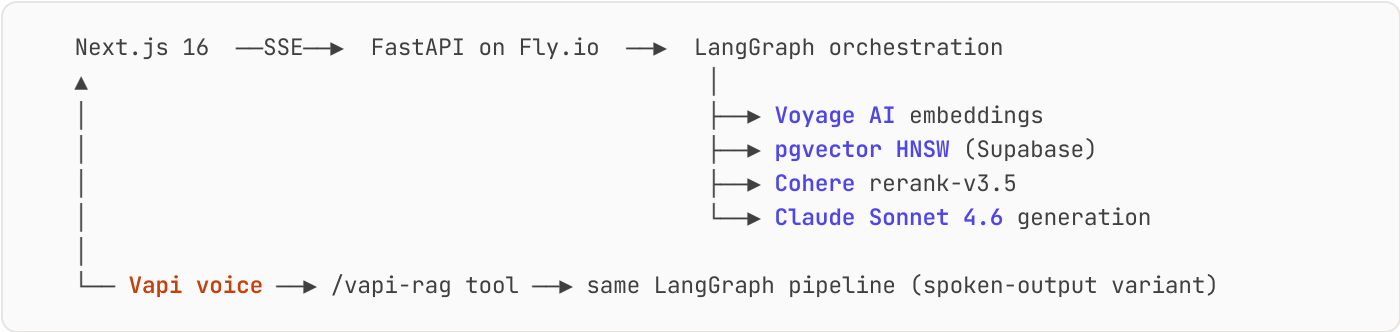
Clinicians need fast, grounded answers from official guidelines (CDC, USPSTF, NHLBI). Hallucinating LLMs are unsafe in this setting — every claim must trace back to a specific page of a real document, and the system must refuse questions outside its evidence base.

Off-the-shelf RAG demos break on this domain because clinical PDFs are layout-heavy (multi-column tables, footnotes, drug-dosing matrices) so naive PDF extraction loses structure and citations; single-vector retrieval misses exact drug names and dosages while pure keyword search misses semantic intent; and a single LLM call has no way to refuse out-of-scope questions or detect when the retrieved evidence doesn't answer the question.

The build (5 days, solo)

DAY	DELIVERABLE	TECH
1	Parse-quality gate over 8 clinical PDFs · 378 pages	Docling (IBM) — layout-aware Markdown
2	Hybrid retrieval over 1,614 chunks · HNSW + GIN tsvector + RRF + rerank	pgvector · Voyage voyage-3-large
3	LangGraph agent — triage → retrieve → grade → refine loop → answer	LangGraph · Claude Sonnet 4.6
4	Premium light-mode UI with SSE token streaming + live pipeline trace	Next.js 16 · Tailwind · Framer Motion
5	Live cloud deploy + Vapi voice mode	Vercel · Fly.io · Supabase · Vapi

Architecture



Every answer chunk is anchored to `source.pdf · page N · heading-path`. The grade node uses a separate Claude call to verify the evidence answers the question before generating; if insufficient, a refine node rewrites the query in clinical terminology and retries (max 2 loops).

Results

STAGE	LOCAL	LIVE (FLY + SUPABASE)
Retrieval (vector + BM25 + Cohere rerank)	~1.0s	~1.2s
Triage + grade (Claude calls)	~1.0s	~1.1s
Answer streaming — first token	~0.6s	~0.8s
Full answer end-to-end	3-5s	3-5s
Cited-section accuracy (10-query smoke test)	10/10 doc · 9/10 section	identical

Why this matters for a clinic or health-tech client

- **HIPAA-aware posture by design** — no PHI in this demo, but the architecture supports identity-gated routing, evidence-only answers, BAA-eligible inference vendors (Anthropic + Voyage), and audit logging of every retrieval.
- **Polyglot stack any team can fork** — Python for the agent layer (where data scientists already live), TypeScript for the UI (where most product teams live).
- **Plugs into voice** — same backend serves Vapi via a custom function tool; a clinician can phone the agent and get the same evidence-grounded answer the UI streams inline.

Stack

Claude Sonnet 4.6

Voyage voyage-3-large

Cohere rerank-v3.5

Docling

LangGraph

PostgreSQL + pgvector (HNSW)

tsvector GIN

FastAPI + SSE

Next.js 16 (App Router)

Tailwind + Framer Motion

Fly.io

Supabase

Vercel

Vapi + Deepgram